

1 Regular Parametric Surface

Definition 1.(Parametric Surface)

A **parametric surface** is a continuous map

$$S : D(\hookrightarrow \mathbb{R}^2) \longrightarrow \mathbb{R}^3$$

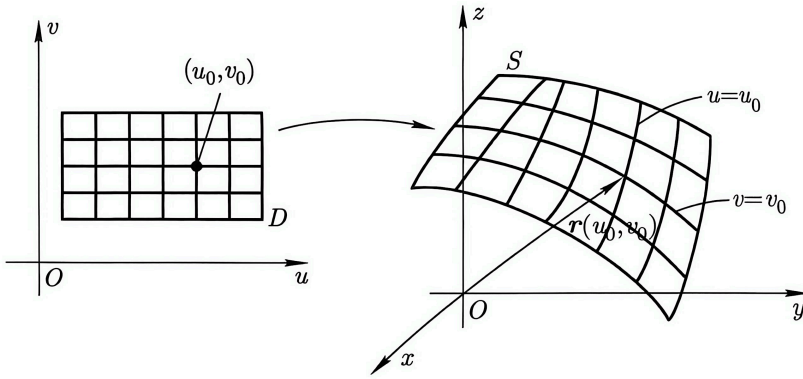
$$(u, v) \mapsto (x(u, v), y(u, v), z(u, v)),$$

where $D \subseteq \mathbb{R}^2$ is a domain, i.e., open and connected subset of $\mathbb{R}^3$.

Remark: We mainly focus on the surface such that $x(u, v), y(u, v), z(u, v) \in C^\infty(\mathbb{R}^2; \mathbb{R})$.

Remark: We also write the parametric surface S in the form $\mathbf{r} = \mathbf{r}(u, v) \triangleq (x(u, v), y(u, v), z(u, v))$, i.e., for a given point $p_0 \in S$, we can write $\overrightarrow{Op_0} \triangleq \mathbf{r}(u_0, v_0)$.

• For a given value $u = u_0$, if we let the parameter v vary, then we get a curve $\mathbf{r}(u_0, v)$ that lies in the surface S , which is called the ***v-curve***, we also have the ***u-curve*** $\mathbf{r}(u, v_0)$ in the same way. The ***u-curve*** and ***v-curve*** together form the **parametric net** of the surface. The parameters (u, v) are called the **curvilinear coordinates** of the surface.



Definition 2.(Regular Parametric Surface)

A parametric surface is called a **regular parametric surface** if it admits a parametrization $\mathbf{r} = \mathbf{r}(u, v)$ such that

$$\partial_u \mathbf{r}|_{(u_0, v_0)} \times \partial_v \mathbf{r}|_{(u_0, v_0)} \neq \mathbf{0}$$

for all points (u_0, v_0) in the parameter domain, where we use the notation that $\partial_u \mathbf{r} := \frac{\partial \mathbf{r}}{\partial u}|_{(u_0, v_0)}$, $\partial_v \mathbf{r} := \frac{\partial \mathbf{r}}{\partial v}|_{(u_0, v_0)}$. (or $\mathbf{r}_u := \frac{\partial \mathbf{r}}{\partial u}$, $\mathbf{r}_v := \frac{\partial \mathbf{r}}{\partial v}$.)

Remark: By this definition we see that the tangent vector $\partial_u \mathbf{r}|_{(u_0, v_0)}$ of the ***u-curve*** $\mathbf{r}(u, v_0)$ and the tangent vector $\partial_v \mathbf{r}|_{(u_0, v_0)}$ of the ***v-curve*** $\mathbf{r}(u_0, v)$ are never parallel to each other at any point $\mathbf{r}(u_0, v_0)$ on the regular parametric surface S .

• Let $S : \mathbf{r} = \mathbf{r}(u, v) \ (u, v) \in D$ be a regular parametric surface, then for any given $(u_0, v_0) \in D$, we have

$$\partial_u \mathbf{r} \times \partial_v \mathbf{r}|_{(u_0, v_0)} = \left(\frac{\partial(y, z)}{\partial(u, v)}, \frac{\partial(z, x)}{\partial(u, v)}, \frac{\partial(x, y)}{\partial(u, v)} \right) \Big|_{(u_0, v_0)} \neq \mathbf{0},$$

where we have used the notation $\frac{\partial(y,z)}{\partial(u,v)} := \det \begin{pmatrix} \frac{\partial y}{\partial u} & \frac{\partial z}{\partial u} \\ \frac{\partial y}{\partial v} & \frac{\partial z}{\partial v} \end{pmatrix}$. Without loss of generality, we may assume that

$$\left. \frac{\partial(x,y)}{\partial(u,v)} \right|_{(u_0,v_0)} \neq 0,$$

by inverse function theorem, we see that there is a neighborhood U of (u_0, v_0) in D such that

$$\begin{cases} u = u(x, y) \\ v = v(x, y) \end{cases}, \quad (x, y) \in U,$$

such that

$$x(u(x, y), v(x, y)) \equiv x, \quad y(u(x, y), v(x, y)) \equiv y, \quad z(u(x, y), v(x, y)) \triangleq z(x, y).$$

Thus the regular parametric surface S can be given by

$$S|_U : \mathbf{r} = \mathbf{r}(x, y) = (x, y, z(x, y)) \quad (x, y) \in U,$$

which shows that there is a bijection

$$\begin{aligned} U &\longleftrightarrow S|_U \\ (x, y) &\longleftrightarrow (x, y, z(x, y)). \end{aligned}$$

• By the discussion above we see that for any point $(u_0, v_0) \in D$, there must be a neighborhood U of (u_0, v_0) in D such that the regular parametric surface can be expressed locally as $S|_U \triangleq (x, y, z(x, y)) \quad (x, y) \in U$, which is called the **Monge form** of the regular parametric surface S . Conversely, any surface S that is given by the Monge form must be a regular parametric surface: Suppose $S : \mathbf{r} = \mathbf{r}(x, y) = (x, y, z(x, y))$, then we get

$$\partial_x \mathbf{r} = (1, 0, \partial_x z), \quad \partial_y \mathbf{r} = (0, 1, \partial_y z),$$

and thus

$$\partial_x \mathbf{r} \times \partial_y \mathbf{r} = (-\partial_x z, -\partial_y z, 1) \neq \mathbf{0}.$$

Definition 3. (Admissible Change of Parameters)

Let $S : \mathbf{r} = \mathbf{r}(u, v)$ be a regular parametric surface, then $(\tilde{u}, \tilde{v})$ is said to be an **admissible change of parameters** if the following two conditions are satisfied:

- (1) $(u, v) = (u(\tilde{u}, \tilde{v}), v(\tilde{u}, \tilde{v}))$, i.e., S can be expressed as $S : \mathbf{r} = \mathbf{r}(\tilde{u}, \tilde{v})$;
- (2) $\frac{\partial(u,v)}{\partial(\tilde{u},\tilde{v})} \neq 0$.

Proposition 1. Let $(\tilde{u}, \tilde{v})$ be an admissible change of parameters of a regular parametric surface $S : \mathbf{r} = \mathbf{r}(u, v)$, then $S : \mathbf{r} = \mathbf{r}(\tilde{u}, \tilde{v})$ is still a regular parametric surface.

Proof: Since $\mathbf{r} = \mathbf{r}(u, v) = \mathbf{r}(u(\tilde{u}, \tilde{v}), v(\tilde{u}, \tilde{v}))$, so we get

$$\partial_{\tilde{u}} \mathbf{r} = \partial_u \mathbf{r} \cdot \partial_{\tilde{u}} u + \partial_v \mathbf{r} \cdot \partial_{\tilde{u}} v,$$

$$\partial_{\tilde{v}}\mathbf{r} = \partial_u\mathbf{r} \cdot \partial_{\tilde{v}}u + \partial_v\mathbf{r} \cdot \partial_{\tilde{v}}v.$$

Hence

$$\begin{aligned} \partial_{\tilde{u}}\mathbf{r} \times \partial_{\tilde{v}}\mathbf{r} &= (\partial_u\mathbf{r} \cdot \partial_{\tilde{u}}u + \partial_v\mathbf{r} \cdot \partial_{\tilde{u}}v) \times (\partial_u\mathbf{r} \cdot \partial_{\tilde{v}}u + \partial_v\mathbf{r} \cdot \partial_{\tilde{v}}v) \\ &= (\partial_{\tilde{u}}v \cdot \partial_{\tilde{v}}u)(\partial_v\mathbf{r} \times \partial_u\mathbf{r}) + (\partial_{\tilde{u}}u \cdot \partial_{\tilde{v}}v)(\partial_u\mathbf{r} \times \partial_v\mathbf{r}) \quad (\text{since } \partial_u\mathbf{r} \times \partial_u\mathbf{r} = \mathbf{0}, \partial_v\mathbf{r} \times \partial_v\mathbf{r} = \mathbf{0}) \\ &= (\partial_{\tilde{u}}u \cdot \partial_{\tilde{v}}v - \partial_{\tilde{u}}v \cdot \partial_{\tilde{v}}u)(\partial_u\mathbf{r} \times \partial_v\mathbf{r}) \\ &= \frac{\partial(u, v)}{\partial(\tilde{u}, \tilde{v})} \cdot (\partial_u\mathbf{r} \times \partial_v\mathbf{r}) \neq \mathbf{0}, \end{aligned}$$

which means that $S : \mathbf{r} = \mathbf{r}(\tilde{u}, \tilde{v})$ is still a parametric surface. ■

Remark: By this proposition we know that the properties of a regular parametric surface are invariant under admissible changes of parameters.

Definition 4.

The side to which the vector $\partial_u\mathbf{r} \times \partial_v\mathbf{r}$ points is defined as the positive side of the surface.

Remark:

(1) By the definition above, the order of the parameters (u, v) determines the orientation of the regular parametric surface.

(2) The admissible change of parameters $(u, v) = (u(\tilde{u}, \tilde{v}), v(\tilde{u}, \tilde{v}))$ preserves the orientation of a regular parametric surface $\iff \frac{\partial(u, v)}{\partial(\tilde{u}, \tilde{v})} > 0$.

• Many surfaces cannot be covered by a single regular parametrization; rather, it must be partitioned into a number of patches, each of which admits a regular parametrization, which result in the following definition.

Definition 5.(Regular Surface)

Let $S \subseteq \mathbb{R}^3$ be a subset of $\mathbb{R}^3$. If for any point $p \in S$, there are neighborhoods $V_p \subseteq \mathbb{R}^3$ and $U_p \subseteq \mathbb{R}^2$ of p such that there is a homeomorphism

$$\varphi_p : V_p \cap S \longrightarrow U_p$$

and the map

$$\begin{aligned} \varphi_p^{-1} = \mathbf{r} : U_p &\longrightarrow V_p \cap S \\ (u, v) &\mapsto (x(u, v), y(u, v), z(u, v)) \end{aligned}$$

is a regular parametric surface, then S is called a **regular surface**.

Convention: If there is no confusion, by **surface** we usually mean a regular surface.

Remark: By this definition we see that a regular surface is obtained by gluing together regular parametric surface patches.

Remark: If S has two different regular parametrization

$$\mathbf{r}_i : U_i \longrightarrow V_i \cap S \quad \text{for } i = 1, 2,$$

such that $V_1 \cap V_2 \cap S \neq \emptyset$, then there must be two curvilinear coordinates (u_1, v_1) and (u_2, v_2) around $p \in V_1 \cap V_2 \cap S$ such that $(u_1, v_1) = (u_1(u_2, v_2), v_1(u_2, v_2))$ is an admissible change of parameters.

Proof: Since

$$\begin{aligned}\mathbf{r}_1 &= \mathbf{r}_1(u_1, v_1) = (x_1(u_1, v_1), y_1(u_1, v_1), z_1(u_1, v_1)), \\ \mathbf{r}_2 &= \mathbf{r}_2(u_2, v_2) = (x_2(u_2, v_2), y_2(u_2, v_2), z_2(u_2, v_2)),\end{aligned}$$

so locally around $p \in V_1 \cap V_2 \cap S$, we have

$$x_1(u_1, v_1) = x_2(u_2, v_2), \quad y_1(u_1, v_1) = y_2(u_2, v_2), \quad z_1(u_1, v_1) = z_2(u_2, v_2).$$

Since $\partial_{u_1} \mathbf{r}_1 \times \partial_{v_1} \mathbf{r}_1 \neq \mathbf{0}$, so w.l.o.g. we may assume that

$$\frac{\partial(x_1, y_1)}{\partial(u_1, v_1)} \neq 0,$$

which implies that there is

$$u_1 \triangleq f(x_1, y_1), \quad v_1 \triangleq g(x_1, y_1),$$

on a neighborhood of (u_p, v_p) such that

$$f(x_1(u_1, v_1), y_1(u_1, v_1)) \equiv u_1, \quad g(x_1(u_1, v_1), y_1(u_1, v_1)) \equiv v_1.$$

Thus locally, we obtain the change of parameters

$$u_1 = f(x_2(u_2, v_2), y_2(u_2, v_2)), \quad v_1 = g(x_2(u_2, v_2), y_2(u_2, v_2)),$$

and by the property of Jacobian determinant we have

$$\frac{\partial(u_1, v_1)}{\partial(u_2, v_2)} = \frac{\partial(f, g)}{\partial(x_1, y_1)} \cdot \frac{\partial(x_2, y_2)}{\partial(u_2, v_2)} = \left(\frac{\partial(x_1, y_1)}{\partial(u_1, v_1)} \right)^{-1} \cdot \frac{\partial(x_2, y_2)}{\partial(u_2, v_2)} \neq 0,$$

which completes the proof. ■

Example 1.(Sphere)

Consider the sphere

$$\mathbb{S}^2 := \{(x, y, z) \in \mathbb{R}^3 \mid x^2 + y^2 + z^2 = r^2\},$$

the parametric equation for the upper hemisphere is given by

$$z = \sqrt{r^2 - x^2 - y^2}, \quad (x, y) \in \mathbb{B}(\mathbf{0}; r) := \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 < r^2\},$$

i.e., the upper hemisphere can be given by

$$S_1 : \mathbf{r}_1(x, y) = (x, y, \sqrt{r^2 - x^2 - y^2}), \quad \forall (x, y) \in \mathbb{B}(\mathbf{0}; r),$$

similarly, the lower hemisphere can be given by

$$S_2 : \mathbf{r}_2(x, y) = (x, y, -\sqrt{r^2 - x^2 - y^2}), \quad \forall (x, y) \in \mathbb{B}(\mathbf{0}; r).$$

So $\mathbf{r}_1(x, y)$ and $\mathbf{r}_2(x, y)$ are both Monge forms and hence regular parametric surfaces. However, the sphere $\mathbb{S}^2$ cannot be obtained by gluing S_1 and S_2 together because we have lost the equator.

Example 2. (Circular Cylinder)

Consider the circular cylinder

$$S = \{(x, y, z) \in \mathbb{R}^3 \mid x^2 + y^2 = a^2\},$$

whose regular parametrization can be given by

$$\mathbf{r} = \mathbf{r}(u, v) := (a \cos u, a \sin u, bv) \quad \text{with } a > 0, b \text{ constant,}$$

and it's easy to see that $\partial_u \mathbf{r} \times \partial_v \mathbf{r} \neq \mathbf{0}$. The regular parametric surfaces

$$S_1 : \mathbf{r}_1(u, v) = (a \cos u, a \sin u, bv) \quad (u, v) \in (-\pi, \pi) \times \mathbb{R},$$

$$S_2 : \mathbf{r}_2(u, v) = (a \cos u, a \sin u, bv) \quad (u, v) \in (0, 2\pi) \times \mathbb{R}$$

are the sets $S \setminus \{(x, y, z) \in \mathbb{R}^3 \mid x = -a, y = 0, z = bv\}$ and $S \setminus \{(x, y, z) \in \mathbb{R}^3 \mid x = a, y = 0, z = bv\}$ respectively, thus the circular cylinder S can be obtained by gluing the regular parametric surfaces S_1 and S_2 together and hence the circular cylinder S is a regular surface.

Example 3. Let us consider the sphere $\mathbb{S}^2 : \{(x, y, z) \in \mathbb{R}^3 \mid x^2 + y^2 + z^2 = r^2\}$ again. If we use the parametric equation

$$\mathbf{r}(\theta, \varphi) := (r \sin \varphi \cos \theta, r \sin \varphi \sin \theta, r \cos \varphi),$$

hence

$$\partial_\theta \mathbf{r} \times \partial_\varphi \mathbf{r} = r^2 \sin \varphi (\sin \varphi \cos \theta, \sin \varphi \sin \theta, \cos \varphi),$$

which means that $\partial_\theta \mathbf{r} \times \partial_\varphi \mathbf{r} = \mathbf{0}$ when $\varphi = 0$ or $\varphi = \pi$, i.e., the north and south poles. Moreover, if we require $\theta \in (0, 2\pi)$, then it omits an entire meridian (the curve $\theta = 0$, which is identified with $\theta = 2\pi$). Therefore, a single spherical coordinate patch cannot cover the whole sphere.

Now, we use two spherical coordinate systems with different polar axes:

(a) Use the standard spherical coordinates with the z -axis as the polar axis. Choose the domain

$$(\theta, \varphi) \in (0, 2\pi) \times (0, \pi).$$

This regular parametric surface covers the sphere except for north and south poles, and one meridian.

(b) Use spherical coordinates with a different polar axis, for example, the x -axis. Again take the domain

$$(\theta', \varphi') \in (0, 2\pi) \times (0, \pi).$$

Since the polar axis of the second regular parametric surface differs from that of the first one, the points omitted by the first regular parametric surface (the north and south poles) are not poles in the second coordinate system; hence they are covered by the second regular parametric surface. Similarly, the meridian omitted by the first regular parametric surface is covered by the second one.

Thus the sphere is a regular surface.